

Condition-Aware Selective Abstention for Bias Benchmark Question Answering: A Multi-Signal and Explanation Audit

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ABSTRACT Large language models can answer social-bias-sensitive questions by relying on demographic stereotypes when context is ambiguous, yet conservative debiasing can also damage utility by abstaining when context provides a clear answer. This paper studies that trade-off in multiple-choice bias benchmark question answering. We introduce and audit Condition-Aware Selective Abstention, a frozen-model inference-time policy that predicts whether a BBQ-style instance is ambiguous or disambiguated and then applies condition-specific abstention without oracle condition labels at test time. We also audit a seven-signal, condition-aware mixture-of-experts (MoE) retention variant using evidence, stability, confidence, prompt-sensitivity, attention, and sparse-autoencoder features. The audit reveals a structural property of the original BBQ split: the ambiguous/disambiguated boundary is highly condition-separable. An embedding-only condition classifier reaches 0.9961 test accuracy, and the full feature classifier reaches 0.9983. Consequently, a condition-only retention rule achieves 1.0000 ambiguous accuracy, 0.8789 disambiguated accuracy, and 0.0726 false-abstention rate, while the full multi-signal MoE variant achieves 0.9946, 0.8732, and 0.0843, respectively. The multi-signal layer is more useful as a fallback under scarce condition supervision: with only 1% condition-label supervision, the hybrid improves disambiguated accuracy from 0.6786 to 0.8247 and reduces false abstention from 0.2931 to 0.1452. We further add deterministic rule-based explanations that separate runtime rationales from benchmark-only audit labels. Leave-one-category-out, Open-BBQ transfer, KoBBQ condition-transfer audits, and Qwen/Mistral cross-model experiments reduce the risk that the result is only original-split overfitting, while also showing that condition prediction becomes harder outside the main split. We therefore frame the contribution as a no-oracle selective-abstention analysis and audit, not as a causal claim that any single signal or MoE component explains the improvement.

INDEX TERMS Bias benchmark question answering, large language models, selective abstention, condition-aware abstention, selective prediction, abstention audit, explainability, fairness evaluation.

I. INTRODUCTION

LARGE language models (LLMs) are increasingly used as general-purpose question-answering systems, including in settings where user queries contain demographic attributes or socially sensitive contexts. Bias evaluation for question answering is difficult because the desired behavior changes with context. In the Bias Benchmark for Question Answering (BBQ) [1], ambiguous contexts should usually lead to an unknown answer, while disambiguated contexts should allow the model to answer from the evidence. A

debiasing method that always abstains may appear safe on ambiguous items but can sharply reduce utility on disambiguated items. Conversely, a method that preserves utility can still propagate stereotypes when the context does not identify the correct answer.

This paper frames bias mitigation in BBQ-style question answering as a condition-aware abstention problem. Instead of changing model weights, we keep the base model frozen and ask whether a deployable system can decide when to preserve the model's original answer and when to override it

to the unknown option. The central component is a condition predictor that estimates whether an instance is ambiguous or disambiguated without using oracle condition labels at test time. We then audit a multi-signal retention model that combines surface evidence checks, counterfactual stability, log-probability confidence, self-consistency, prompt sensitivity, bias-head attention, and sparse-autoencoder (SAE) feature activation. Finally, we attach a rule-based explanation layer that states whether a decision abstained because the context was predicted ambiguous, preserved utility in a predicted-disambiguated context, or blocked a raw answer that was stereotyped under the BBQ metadata. The intended contribution is system-level and diagnostic: we study how much of the observed gain comes from condition separability versus the seven-signal MoE retention model, rather than claiming that any single signal is a complete mechanistic explanation of bias.

The contributions are as follows:

- We introduce and audit a deployment-oriented frozen-model selective-abstention framework for BBQ-style multiple-choice QA, isolating the contribution of condition prediction from multi-signal retention.
- We evaluate a deployable predicted-condition policy that does not use oracle ambiguity labels at test time, including a simple condition-only retention rule and a seven-signal MoE retention variant.
- We show that the multi-signal layer is most useful as a fallback when condition supervision is limited or uncertain, improving low-label utility and false-abstention behavior relative to a pure condition-only rule.
- We add a deterministic explanation layer that reports runtime rationales and separate benchmark audit labels for blocked stereotypes, residual bias slips, false abstention, and utility-preserving keeps.
- We provide clean five-seed comparisons against Composite, DeCAP [2], and a self-debiasing-style reprompting baseline inspired by [3], with matched-ID coverage and paired bootstrap reporting.
- We add robustness and audit experiments: condition-classifier feature ablations, leave-one-category-out (LOCO), Open-BBQ transfer, cross-LLM robustness, threshold audits, signal-subset ablations, low-label hybrid audits, residual ambiguous-bias count analysis, and a limited-overlap FairSteer [4] appendix check.

The main claim is not that every bias metric is uniformly best or that the MoE is the sole causal driver. Rather, BBQ-style condition prediction is highly informative, and condition-aware selective abstention can maintain high ambiguous abstention accuracy while preserving disambiguated utility under a no-oracle test setting.

II. RELATED WORK

A. BIAS BENCHMARKS FOR QUESTION ANSWERING

BBQ [1] evaluates social bias in multiple-choice question answering across demographic categories. It is especially useful

because it separates ambiguous contexts, where the correct answer is unknown, from disambiguated contexts, where the evidence identifies a target answer. This separation exposes a key failure mode: methods can reduce biased selections by overusing abstention.

B. PROMPT-BASED AND CONTEXT-ADAPTIVE DEBIASING

Prompt-based debiasing methods attempt to reduce stereotype-driven outputs using instructions, self-reflection, or generated debiasing prompts. Self-debiasing [3] uses the model's own zero-shot recognition and correction ability, while DeCAP [2] generates context-adaptive debiasing guidance for zero-shot QA. These approaches are attractive because they do not require weight updates, but prompt sensitivity and false abstention remain practical concerns. Our experiments use only a two-prompt recognition-and-correction baseline inspired by self-debiasing, not an official reproduction of the full protocol in [3].

C. SELECTIVE PREDICTION, ABSTENTION, AND CALIBRATION

The proposed override mechanism is related to reject-option classification and selective prediction. Classical reject-option work studies the trade-off between recognition error and rejection probability [5], while modern selective classification builds abstaining neural classifiers by trading coverage for lower risk [6]. Confidence calibration is also relevant because raw neural confidence scores are often miscalibrated [7]. Our setting differs from standard selective classification because the abstention action has asymmetric semantics: abstention is correct for ambiguous BBQ examples but is an error on disambiguated examples. This motivates condition-aware thresholding rather than a single global confidence cutoff.

D. REPRESENTATION-LEVEL SIGNALS AND STEERING

Activation and representation methods, including dynamic steering approaches such as FairSteer [4], aim to identify and intervene on bias-related internal directions. Sparse autoencoders have also become useful interpretability tools for extracting sparse features from LLM activations [8], [9]. Our method uses representation-level signals conservatively: SAE activation is included as one audited feature source, but we do not claim that the SAE signal alone causes the observed improvements.

E. OTHER BIAS EVALUATION BENCHMARKS

Other benchmarks such as StereoSet [10], CrowS-Pairs [11], and WinoGender [12] evaluate stereotype association or gendered coreference behavior. These datasets are useful external checks, but their task semantics differ from BBQ because they do not always contain an explicit unknown option or the same ambiguous-versus-disambiguated decision boundary. We therefore keep non-BBQ transfer results as robustness evidence rather than the main basis for our claims.

III. TASK AND METRICS

Each example contains a context x , question q , answer choices $A = \{a_1, a_2, a_3\}$, a gold answer y , and a condition $c \in \{\text{ambiguous}, \text{disambiguated}\}$. In ambiguous examples, the correct behavior is to choose the unknown option. In disambiguated examples, the correct behavior is to choose the context-supported non-unknown answer.

We report three main metrics. Ambiguous accuracy Acc_{amb} is the accuracy on ambiguous examples, where correct behavior is abstention. Disambiguated accuracy Acc_{dis} is the accuracy on disambiguated examples, where correct behavior is utility-preserving answer selection. False-abstention rate FAR is the fraction of disambiguated examples where the model incorrectly predicts unknown. We also compute residual ambiguous bias scores on non-unknown ambiguous predictions, but we treat them as diagnostic rather than primary because their denominator can become very small when Acc_{amb} is near one.

IV. METHOD

Fig. 1 summarizes the proposed pipeline. The base LLM is frozen. For each instance, we first run four prompt views: a vanilla QA prompt, a debiasing instruction prompt, a chain-of-thought prompt, and a counterfactual-swap prompt. We then extract seven scalar or low-dimensional signals and a question embedding. The deployable policy first predicts whether the example is ambiguous or disambiguated. A simple condition-only version directly uses that prediction to abstain on ambiguous examples and preserve the primary answer on disambiguated examples. A multi-signal MoE version additionally predicts an answer-retention score before applying condition-specific thresholds.

A. MULTI-PROMPT SIGNAL EXTRACTION

Table 1 lists the seven audited signals. The signals are intentionally heterogeneous: some check whether the answer is supported by context, some check stability under prompt and demographic perturbations, and some use internal model behavior. The later ablations test whether these signals are necessary on the clean BBQ split.

The signal set is organized around three practical failure modes. Evidence and confidence signals target unsupported or low-certainty answers. Counterfactual, consistency, and prompt-sensitivity signals target unstable answers whose prediction changes under small semantic or prompting perturbations. Attention-head and SAE signals provide internal diagnostic features for cases where surface evidence alone may miss stereotype reliance. The method does not require each signal to be independently sufficient; instead, the MoE variant learns a validation-calibrated retention policy from their joint pattern.

B. CONDITION-AWARE ABSTENTION AND MOE RETENTION

Let $s_i \in \mathbb{R}^7$ be the extracted signal vector for example i , and let e_i be a question embedding. A lightweight condition

classifier predicts whether the instance is ambiguous or disambiguated. A four-expert MoE then maps the signal vector to a confidence score:

$$g_i = \text{softmax}(W_g e_i + b_g), \quad (1)$$

$$r_i = \sum_{k=1}^4 g_{i,k} f_k(s_i, e_i), \quad (2)$$

where f_k is the k -th expert and r_i is the final answer-retention score. If r_i falls below a condition-specific threshold, the answer is overridden to unknown; otherwise the original selected answer is retained. The deployable setting uses the predicted condition, while the oracle condition is reported only as an upper bound.

We also evaluate a condition-only retention rule:

$$\tilde{y}_i = \begin{cases} \text{unknown}, & \hat{c}_i = \text{ambiguous}, \\ \hat{y}_i, & \hat{c}_i = \text{disambiguated}, \end{cases} \quad (3)$$

where $\hat{c}_i$ is the predicted condition and $\hat{y}_i$ is the primary answer from the frozen LLM. This rule is intentionally simple. It tests whether the benchmark's ambiguous/disambiguated boundary is sufficiently encoded in the available features to explain the main gains without relying on the MoE retention score.

The MoE is trained to estimate whether the primary answer should be retained. For example i , the target is $z_i = \mathbb{I}[\hat{y}_i = y_i]$, where $\hat{y}_i$ is the original selected answer before override. The training objective combines binary cross-entropy on z_i , a bias penalty that discourages retaining stereotyped non-unknown answers in ambiguous contexts, and a load-balancing term that discourages expert collapse:

$$\mathcal{L} = \mathcal{L}_{BCE} + \lambda_{bias} \mathcal{L}_{bias} + \lambda_{lb} \mathcal{L}_{lb}. \quad (4)$$

In the clean experiments, the default weights are $\lambda_{bias} = 0.5$ and $\lambda_{lb} = 0.1$. Thresholds are selected only on validation data. The oracle-condition thresholds are included to quantify an upper bound; the main deployable variant applies the same thresholding rule with predicted conditions.

C. HYBRID FALLBACK AND RULE-BASED EXPLANATIONS

The clean split shows that condition-only abstention is already close to a ceiling. We therefore use the seven-signal retention model as an optional fallback rather than treating it as universally necessary. The hybrid policy first checks the condition classifier confidence, defined as the larger of the logistic-regression probabilities for ambiguous and disambiguated. A confidence threshold and a MoE risk threshold are selected only on validation data. If confidence is above the selected threshold, the condition-only rule is used; otherwise, the policy falls back to the MoE retention score. This design tests whether the additional signals help most when condition prediction is weak, rather than forcing them into every decision.

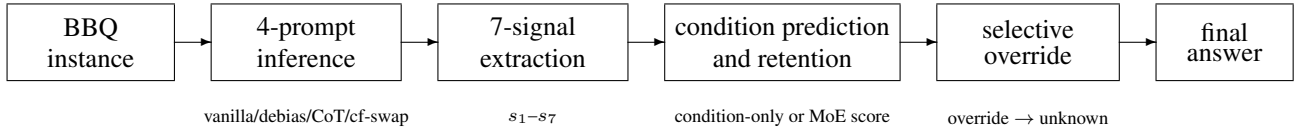


FIGURE 1. Frozen-model pipeline with prompt views, signal extraction, condition prediction, and selective override.

TABLE 1. Signals used by the proposed method. The representation-level signal s_7 is included as an audited auxiliary feature source, not as a standalone causal explanation.

Signal	Code name	Measurement	Interpretation
s_1	s1_evidence	Whether a context quote supports the selected answer	Weak evidence suggests an override candidate.
s_2	s2_counterfactual	Whether the answer remains stable under demographic group swaps	Sensitivity to group identity suggests possible bias reliance.
s_3	s3_confidence	Choice-level log-probability confidence	Low confidence supports abstention.
s_4	s4_consistency	Agreement under repeated stochastic sampling	Inconsistent answers are less reliable.
s_5	s5_bias_head	Attention mass from identified bias-relevant heads to demographic tokens	High demographic attention may indicate stereotype reliance.
s_6	s6_prompt	Agreement across vanilla, debiasing, CoT, and counterfactual prompts	Prompt-sensitive answers are lower-confidence.
s_7	s7_sae_feature	Activation of selected Llama-Scope SAE bias features	Audited representation-level bias signal.

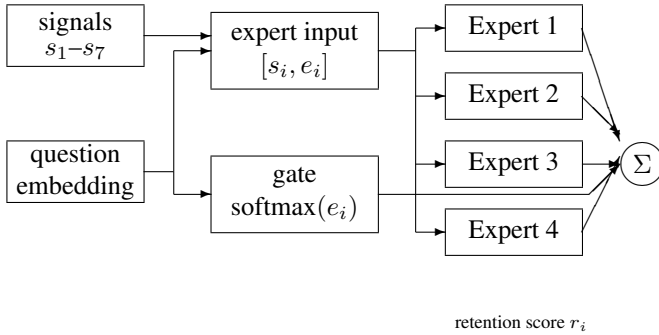


FIGURE 2. Condition-aware MoE retention model. The gate uses the question embedding, while the experts receive the signal vector and embedding before weighted summation.

We also generate deterministic explanations for each final decision. Runtime explanations use only deployable information: the predicted condition, the retained or overridden answer, and thresholded signal flags such as weak context evidence, counterfactual instability, low choice confidence, prompt sensitivity, high demographic attention, or active SAE bias features. Separately, benchmark audit explanations use gold BBQ metadata after evaluation to label decision types such as blocked stereotyped raw answer, residual bias slip, false abstention, or utility-preserving keep. This separation avoids using gold labels at deployment while still making the benchmark behavior inspectable.

V. EXPERIMENTAL SETUP

A. MODELS AND DATA

The main experiments use Llama-3.1-8B [13] on BBQ. Robustness experiments use Qwen-2.5-7B [14] and Mistral-7B-v0.3, with Mistral cited through the Mistral 7B technical report [15]. The clean main evaluation uses five seeds, 42, 123, 456, 789, and 999, with matched test IDs where possible.

For each clean seed, the saved-signal pool is split into 6,208 training examples, 1,328 validation examples, and 1,328 test examples. The MoE is trained for 30 epochs with batch size 32, AdamW, learning rate 10^{-3} , weight decay 10^{-5} , gradient clipping at 1.0, and best-checkpoint selection by validation loss. The condition classifier is a balanced logistic-regression model over category, primary-answer, embedding, and signal features; the primary-answer feature is the model's selected answer, not the gold answer. Its mean clean test accuracy across the five seeds is 0.9983, so the predicted-condition and oracle-condition variants are expected to be close on the main BBQ split. Appendix B shows that removing the primary-answer feature barely changes this result, while embedding-only prediction remains very strong. LOCO is reported separately because condition prediction is harder when a category is held out.

Thresholds are tuned on the validation split rather than the test split. The single-threshold variant searches one retention threshold over a grid, while the condition-aware variants search separate ambiguous and disambiguated thresholds on validation data. The final reported test metrics are then computed once on the held-out test split.

B. BASELINES

We compare against a Composite prompt-only baseline, DeCAP [2], a self-debiasing-style reprompting baseline (SDR),

a primary-answer-only audit baseline, the condition-only retention rule, a single-threshold version of the MoE, and the full predicted-condition MoE. SDR follows the recognition-and-correction idea of [3], but it is our reprompting implementation rather than an official reproduction; because it over-abstains heavily, we treat it as an auxiliary replication note rather than a primary claim. FairSteer is reported only as a limited-overlap auxiliary comparison because the matched-ID overlap is small, with approximately 15 examples per seed.

C. STATISTICAL TESTING AND AUDITS

For main comparisons, we use paired bootstrap tests on matched IDs. We also report LOCO generalization, Open-BBQ zero-shot transfer, cross-LLM robustness, low-threshold audits, condition-classifier ablations, signal-subset ablations, low-label hybrid fallback audits, rule-based explanation audits, and residual ambiguous-bias counts. These analyses test whether the observed gains are driven by category memorization, split overfitting, threshold artifacts, condition-label separability, unstable residual-bias metrics, or a lack of auditable decision rationales.

VI. RESULTS

A. MAIN CLEAN BBQ COMPARISON

Table 2 shows the clean five-seed BBQ results. The strongest deployable result on the main split is the condition-only retention rule, which achieves $Acc_{amb} = 1.0000 \pm 0.0000$, $Acc_{dis} = 0.8789 \pm 0.0070$, and $FAR = 0.0726 \pm 0.0067$. The full seven-signal MoE predicted-condition variant remains strong at $Acc_{amb} = 0.9946 \pm 0.0054$, $Acc_{dis} = 0.8732 \pm 0.0108$, and $FAR = 0.0843 \pm 0.0193$, but it is not the simplest or strongest explanation of the main-split gain. A separately audited hybrid fallback reaches $Acc_{amb} = 0.9979 \pm 0.0020$, $Acc_{dis} = 0.8795 \pm 0.0070$, and $FAR = 0.0723 \pm 0.0071$ when full condition supervision is available, indicating that its main advantage is not the already-easy clean split but the low-label setting analyzed below.

The primary-answer-only audit is also important: it already reaches $Acc_{dis} = 0.8798$ and $FAR = 0.0717$, so the main clean-split gain is asymmetric. The policy mainly converts unsupported ambiguous-context answers into the correct unknown option while preserving, rather than improving, the base model's disambiguated utility. Compared with Composite and DeCAP, the condition-aware policies improve ambiguous accuracy and disambiguated utility while reducing false abstention. Compared with self-debiasing-style reprompting (SDR), they avoid severe over-abstention on disambiguated examples.

FairSteer is excluded from the main table because its matched-ID overlap is too small for a full-coverage comparison; it is reported only in Appendix A. Paired bootstrap tests show strong improvements over Composite and DeCAP on Acc_{amb} , Acc_{dis} , and FAR (maximum observed $p < 0.001$ in the matched tests). These $p < 0.001$ results remain below a conservative Bonferroni threshold for the 12 baseline-

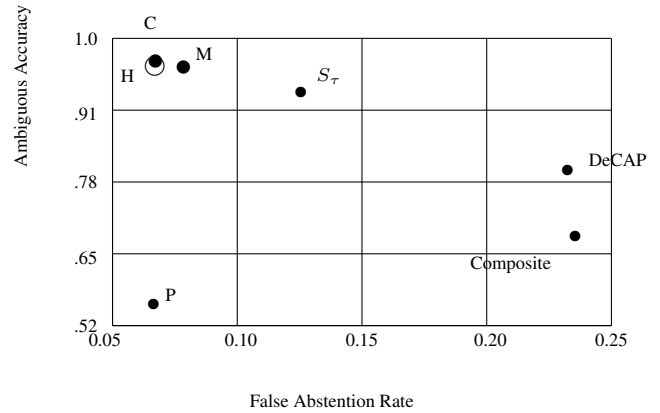


FIGURE 3. Operational trade-off between false abstention and ambiguous accuracy. Point labels: P primary-answer-only, C condition-only, H hybrid, M predicted-condition MoE, and S_T single-threshold MoE. SDR is omitted because its FAR is 0.7858, outside the operational range shown here.

metric comparisons considered in the paired audit; weaker comparisons such as the SDR ambiguous-accuracy delta are not used as primary claims. Against SDR, the safest claim is improved disambiguated utility and FAR rather than a large ambiguous-accuracy gain, because that baseline already abstains frequently on ambiguous examples.

B. WHEN THE SIGNAL FALLBACK HELPS

The clean main split makes condition prediction unusually easy, so the condition-only rule is difficult to improve. To test whether the seven-signal fallback is useful under weaker supervision, we retrain the condition classifier with only a fraction of the condition-labeled training examples and keep the same validation-only hybrid tuning procedure. Table 3 shows that the hybrid fallback is most valuable when condition supervision is scarce, especially at 1–5% labels. With only 1% of condition labels, the hybrid improves Acc_{amb} from 0.9136 to 0.9530, Acc_{dis} from 0.6786 to 0.8247, and reduces FAR from 0.2931 to 0.1452. It also reduces seed-level variance in Acc_{dis} from 0.0674 to 0.0313. By 10% and 100% labels, the hybrid provides only marginal Acc_{dis} or FAR gains and can slightly reduce Acc_{amb} (0.9852 to 0.9834 at 10%; 1.0000 to 0.9979 at 100%). This supports the narrower claim that signals are useful as an uncertainty fallback, not as a universal improvement over condition-only abstention.

C. GENERALIZATION BEYOND THE MAIN SPLIT

Table 4 reports generalization and robustness experiments. In LOCO, each BBQ category is held out from training and used for evaluation. The predicted-condition variant remains strong, with $Acc_{amb} = 0.9214 \pm 0.0421$, $Acc_{dis} = 0.8331 \pm 0.0793$, and $FAR = 0.1161 \pm 0.0551$ across 45 folds. This supports the interpretation that the method is not merely memorizing category-specific patterns.

Open-BBQ transfer provides an external BBQ-style check. On 3,300 Open-BBQ examples, the method achieves

TABLE 2. Clean BBQ main comparison with five seeds using validation-selected operating points. Main claims use Acc_{amb} , Acc_{dis} , and FAR; residual ambiguous bias is analyzed separately because its denominator is often near zero. The condition-only row is an audit baseline that uses predicted condition without the MoE retention score.

System	Mean n	Acc_{amb}	Acc_{dis}	FAR	Role
Composite	1328	0.6843 ± 0.0138	0.2855 ± 0.0109	0.2449 ± 0.0164	Prompt-only baseline
DeCAP	1328	0.8057 ± 0.0055	0.7238 ± 0.0075	0.2419 ± 0.0094	Context-adaptive prompt baseline
SDR (self-debiasing-style)	1328	0.9584 ± 0.0078	0.1928 ± 0.0111	0.7858 ± 0.0083	High over-abstention
Primary answer only	1328	0.5596 ± 0.0152	0.8798 ± 0.0076	0.0717 ± 0.0069	No abstention audit
Condition-only retention	1328	1.0000 ± 0.0000	0.8789 ± 0.0070	0.0726 ± 0.0067	Best deployable audit
Hybrid fallback	1328	0.9979 ± 0.0020	0.8795 ± 0.0070	0.0723 ± 0.0071	Uncertainty fallback audit
MoE, single threshold	1328	0.9494 ± 0.0126	0.8413 ± 0.0184	0.1325 ± 0.0240	MoE without condition split
MoE, predicted condition	1328	0.9946 ± 0.0054	0.8732 ± 0.0108	0.0843 ± 0.0193	Full multi-signal variant
MoE, oracle condition	1328	0.9946 ± 0.0054	0.8738 ± 0.0109	0.0837 ± 0.0194	Upper bound only

TABLE 3. Low-label condition-classifier audit. The hybrid uses the MoE retention score only when the condition classifier is uncertain.

Label frac.	System	Acc_{amb}	Acc_{dis}	FAR
1%	Condition-only	0.9136 ± 0.0051	0.6786 ± 0.0674	0.2931 ± 0.0680
1%	Hybrid fallback	0.9530 ± 0.0054	0.8247 ± 0.0313	0.1452 ± 0.0339
5%	Condition-only	0.9645 ± 0.0097	0.8301 ± 0.0096	0.1280 ± 0.0114
5%	Hybrid fallback	0.9744 ± 0.0110	0.8548 ± 0.0071	0.1048 ± 0.0079
10%	Condition-only	0.9852 ± 0.0034	0.8587 ± 0.0133	0.0958 ± 0.0133
10%	Hybrid fallback	0.9834 ± 0.0028	0.8726 ± 0.0053	0.0831 ± 0.0079
100%	Condition-only	1.0000 ± 0.0000	0.8789 ± 0.0070	0.0726 ± 0.0067
100%	Hybrid fallback	0.9979 ± 0.0020	0.8795 ± 0.0070	0.0723 ± 0.0071

$Acc_{amb} = 0.9915$, $Acc_{dis} = 0.8358$, and $FAR = 0.1012$. Cross-LLM experiments show the same qualitative pattern on Qwen and Mistral, though with higher false-abstention rates than Llama. For Qwen and Mistral, no compatible public SAE feature set was used; the s_7 field is therefore zero/default in those runs, which may contribute to the weaker FAR relative to Llama. Appendix B further audits KoBBQ condition prediction. KoBBQ itself is highly separable when trained and tested within KoBBQ (0.9995 full-feature accuracy), but an English-BBQ-trained embedding classifier does not transfer directly to Korean examples (0.5000 full-feature transfer accuracy). This dissociates benchmark structure from transferred representation: the KoBBQ end-to-end drop is better interpreted as cross-lingual representation and policy-transfer failure than as evidence against condition-aware abstention itself.

D. RISK-COVERAGE AND THRESHOLD BEHAVIOR

The disambiguated retention threshold τ_{dis} controls when a predicted-disambiguated example is kept as the primary answer rather than overridden to unknown. Lower values are less conservative and therefore tend to reduce false abstention. Table 5 shows a stable low-threshold region: $\tau_{dis} \in [0.00, 0.05]$ preserves the same ambiguous accuracy while changing disambiguated accuracy and FAR only modestly. This directly addresses the concern that the selected deployment point is an unstable grid-boundary artifact.

E. INTERPRETABILITY AND ABLATIONS

The MoE routing analysis indicates that expert usage is not strongly dominated by a single category. Table 6 reports

compact routing statistics: mean category Gini is 0.078, normalized entropy is 0.990, and mutual information with category is 0.0178 bits. This weakens the concern that the gate simply memorizes social categories.

Signal ablations show that the seven-signal MoE should not be overinterpreted as the sole driver of the main result. Single-signal masking produces relatively small metric changes in the clean five-seed setting, and the component audit in Appendix D shows that all-zero, s_3 -only, and four-signal MoE variants all remain competitive. In those reduced variants, the disambiguated keep/override decisions match the condition-only rule, which explains the identical Acc_{dis} and FAR values in Table 14. This points to condition prediction and condition-specific abstention as the dominant factor on the original BBQ split. SAE layer comparison selected layer 15 and identified 56 bias-related SAE features, but the observed SAE-only loss deltas are small. Accordingly, we treat s_7 as an audited auxiliary signal rather than the primary causal driver.

F. RULE-BASED EXPLANATION AUDIT

The explanation layer converts each final decision into a deterministic runtime rationale and a separate benchmark audit label. Across the five clean test splits for the predicted-condition variant, it identifies 995 cases where a raw stereotyped answer in an ambiguous context was blocked by abstention, 449 cases where a raw anti-stereotyped but still unsupported answer was blocked, and only 8 residual stereotyped bias slips out of 6,640 seed-level decisions. It also separates 280 false abstentions from 2,899 utility-preserving disambiguated keeps. These counts are not an additional accuracy metric, but they make the decision behavior easier to inspect and help prevent overstating the SAE or MoE components as causal explanations. Appendix F reports the full decision taxonomy.

G. RESIDUAL AMBIGUOUS BIAS

The predicted-condition variant has a residual ambiguous-bias score of 0.8333 ± 0.3333 , which is numerically unstable. This happens because ambiguous accuracy is near perfect: across the five clean seeds, the residual non-unknown ambiguous counts are only 0, 1, 3, 5, and 9. Therefore,

TABLE 4. Generalization and robustness experiments. Except for the LOCO single-threshold row, rows use the predicted-condition variant; the single-threshold row removes condition-specific thresholds.

Experiment	Folds/seeds	Acc_{amb}	Acc_{dis}	FAR
LOCO, predicted condition	45 folds	0.9214 ± 0.0421	0.8331 ± 0.0793	0.1161 ± 0.0551
LOCO, single threshold	45 folds	0.8362 ± 0.0536	0.8013 ± 0.0936	0.1536 ± 0.0777
Open-BBQ transfer	$n = 3300$	0.9915	0.8358	0.1012
Qwen-2.5-7B cross-LLM	5 seeds	0.9895 ± 0.0046	0.8147 ± 0.0183	0.1672 ± 0.0222
Mistral-7B-v0.3 cross-LLM	5 seeds	0.9940 ± 0.0018	0.7798 ± 0.0099	0.1916 ± 0.0100

TABLE 5. Low-threshold plateau for the MoE predicted-condition variant. This is an audit sweep around the low-threshold region, not a separate test-set tuning procedure.

τ_{dis}	Acc_{amb}	Acc_{dis}	FAR
0.00	0.9946 ± 0.0054	0.8798 ± 0.0076	0.0717 ± 0.0069
0.01	0.9946 ± 0.0054	0.8789 ± 0.0087	0.0744 ± 0.0106
0.02	0.9946 ± 0.0054	0.8780 ± 0.0101	0.0771 ± 0.0139
0.03	0.9946 ± 0.0054	0.8768 ± 0.0098	0.0792 ± 0.0152
0.05	0.9946 ± 0.0054	0.8738 ± 0.0109	0.0837 ± 0.0194
0.10	0.9946 ± 0.0054	0.8696 ± 0.0132	0.0916 ± 0.0215

TABLE 6. Category-level MoE routing audit. Lower Gini and higher normalized entropy indicate broadly distributed expert usage.

Routing statistic	Value
Mean category Gini	0.078
Mean normalized entropy	0.990
Mutual information with category	0.0178 bits
Normalized mutual information	0.0089

the denominator for a residual stereotype-vs-anti-stereotype score is too small for a strong “lowest bias score” claim. We report raw counts and confidence intervals in the appendix and use Acc_{amb} , Acc_{dis} , and FAR as the main comparison metrics.

VII. DISCUSSION

The central result is a favorable utility-abstention trade-off produced by condition-aware selective abstention. The clean-split contribution is asymmetric. The primary-answer-only baseline already performs well on disambiguated examples, so the selective policy mainly improves ambiguous-context behavior while preserving disambiguated utility. This is still practically important because a system that answers ambiguous examples with demographic guesses can look useful while failing the intended benchmark semantics. At the same time, the audit shows that the original BBQ split makes condition prediction unusually easy. The predicted-condition and oracle-condition variants are nearly identical because the ambiguous/disambiguated boundary is strongly encoded in embeddings and related surface features, not because the system secretly uses oracle labels.

The auxiliary experiments also clarify what should not be overclaimed. FairSteer is not a main baseline in our analysis because the matched-ID overlap is too small. The SAE signal is included and audited, but the ablation does not justify claiming that SAE feature s_7 is the main reason for

the improvement. The MoE itself is also not the strongest explanation on the clean main split, since the condition-only retention rule performs at least as well. Similarly, residual ambiguous-bias scores should be interpreted with raw residual counts because near-perfect ambiguous abstention leaves too few non-unknown cases for a stable score.

A. NOVELTY AND SCOPE

The novelty claim is intentionally framed as a deployable decision analysis rather than as a single new representation primitive. The method combines condition prediction, heterogeneous signal extraction, optional MoE retention, condition-specific thresholding, and deterministic decision explanations into a no-oracle inference-time abstention policy. This scope matters for evaluation: the relevant question is whether condition-aware abstention improves the utility-abstention trade-off while remaining robust to category hold-out, transfer, and backbone changes, and whether the resulting decisions can be audited after the fact. We do not claim universal bias mitigation across all fairness datasets, a causal mechanistic proof for any individual signal, or that the seven-signal MoE is necessary on the original BBQ split.

VIII. LIMITATIONS

First, the method is designed for BBQ-style multiple-choice QA with an explicit unknown option. Transfer to tasks with different output semantics is less clean. StereoSet, WinoGender, and KoBBQ results are informative but mixed, and should be treated as appendix robustness checks rather than main evidence. Second, the clean BBQ split is highly condition-separable, so results on that split should be interpreted partly as an audit of the benchmark structure. LOCO lowers condition-prediction accuracy, and KoBBQ shows that condition separability can be strong within a new language while still failing under direct English-to-Korean embedding transfer. Third, the multi-prompt signal pipeline increases inference cost compared with a single prompt, although the condition-only audit shows that a simpler deployment policy can be competitive. Fourth, the internal attention-head and SAE signals are diagnostic and should not be interpreted as full mechanistic explanations. Fifth, the rule-based explanations are deterministic audit rationales, not human-validated causal explanations of model cognition. Sixth, all reported results are benchmark-based; human-centered fairness evaluation remains necessary before deployment in real-world decision-support settings.

IX. CONCLUSION

This paper presented Condition-Aware Selective Abstention, a frozen-model framework for bias-sensitive question answering with no-oracle condition-aware abstention. On clean BBQ evaluations, condition-aware retention converts unsupported ambiguous-context answers to unknown while preserving disambiguated utility. The strongest conclusion is that BBQ-style bias evaluation rewards condition-aware selective abstention more than mechanism-aware multi-signal retention on the original split, and that this behavior reflects a structural property of the benchmark. The seven-signal layer remains useful as a fallback when condition supervision is scarce or uncertain, and the rule-based explanation artifacts make blocked stereotypes, residual slips, false abstentions, and utility-preserving keeps inspectable without relying on free-form rationales. LOCO, Open-BBQ transfer, KoBBQ condition-transfer audits, and cross-LLM experiments provide additional evidence beyond the main split, while residual bias scores, SAE-feature interpretations, and MoE-causality claims should be reported cautiously.

APPENDIX A BASELINE COVERAGE AND PAIRED TESTS

Table 7 reports matched-ID coverage for baselines. Composite, DeCAP, and the current SDR full run have full same-ID coverage. FairSteer is not used as a main baseline because its matched overlap is only about 15 examples per seed.

TABLE 7. Matched-ID baseline coverage over five seeds. The SDR legacy subset is retained only for paired-test continuity.

System	Mean n	Min shared n	Max missing
Composite	1328.0	1328.0	0.0
DeCAP	1328.0	1328.0	0.0
SDR, legacy matched subset	187.2	180.0	1148.0
SDR, full run	1328.0	1328.0	0.0
FairSteer	14.8	10.0	1318.0
MoE, predicted condition	1328.0	–	–

Table 8 summarizes the matched paired-bootstrap comparisons available from the clean experiment artifact. Positive accuracy deltas indicate higher accuracy for the MoE predicted-condition variant; negative FAR deltas indicate lower false abstention. These tests support strong claims against Composite and DeCAP. The SDR row uses the earlier 187-ID matched subset and is kept only as a paired statistical check; the full-coverage SDR metrics in Table 2 show the same qualitative over-abstention pattern.

The SDR baseline uses two prompts: an initial multiple-choice prompt that asks for only (A), (B), or (C), followed by a review prompt asking the model to revise the previous answer if it relied on stereotypes and to select the uncertainty option when context is insufficient. This protocol follows the general self-recognition and self-correction idea of self-debiasing, but it is not an official reproduction of [3]. Because the run over-abstains on disambiguated examples, we report it transparently but do not use it as the primary evidence for superiority.

TABLE 8. Paired-bootstrap summary for the predicted-condition variant. Parentheses report the maximum seed-level paired-bootstrap p value; the Bonferroni threshold for 12 baseline-metric tests is $\alpha = 0.0042$.

Baseline	ΔA_{amb}	ΔA_{dis}	ΔFAR
Composite	+0.310 (< 0.001)	+0.588 (< 0.001)	−0.161 (< 0.001)
DeCAP	+0.189 (< 0.001)	+0.149 (< 0.001)	−0.158 (< 0.001)
SDR legacy subset	+0.038 (.161)	+0.662 (< 0.001)	−0.690 (< 0.001)
FairSteer	+0.397 (.120)	0.000 (1.000)	0.000 (1.000)

APPENDIX B CONDITION-PREDICTION AUDIT

Table 9 audits the condition classifier used by the no-oracle variants. The primary-answer feature is not the driver: removing it changes accuracy from 0.9983 to 0.9982. Embedding-only prediction remains 0.9961 on the clean split, indicating that the BBQ ambiguous/disambiguated boundary is strongly encoded in the input representation. In LOCO, the same boundary is less separable, with embedding-only accuracy dropping to 0.8909 and the full feature set to 0.9261.

TABLE 9. Condition-classifier feature ablation over five seeds.

Feature set	Test accuracy
Signals + embedding + category + primary	0.9983 $\pm$ 0.0011
Minus primary answer	0.9982 $\pm$ 0.0014
Signals + embedding	0.9980 $\pm$ 0.0011
Embedding only	0.9961 $\pm$ 0.0023
Signals only	0.8334 $\pm$ 0.0078
Primary answer only	0.5440 $\pm$ 0.0093
Category only	0.5000 $\pm$ 0.0000

TABLE 10. LOCO condition-prediction accuracy over held-out categories.

Feature set	Held-out accuracy
Signals + embedding + primary	0.9261 $\pm$ 0.0273
Signals + embedding	0.9259 $\pm$ 0.0268
Embedding only	0.8909 $\pm$ 0.0320
Signals only	0.8125 $\pm$ 0.0667

TABLE 11. KoBBQ condition-prediction audit. English BBQ $\rightarrow$ KoBBQ trains on English BBQ and evaluates on KoBBQ; Within KoBBQ trains and evaluates within KoBBQ splits, both using the same embedding model.

Scope	Full features	Embedding only	Signals only
English BBQ $\rightarrow$ KoBBQ	0.5000 $\pm$ 0.0000	0.5000 $\pm$ 0.0000	0.6534 $\pm$ 0.0018
Within KoBBQ	0.9995 $\pm$ 0.0011	0.9990 $\pm$ 0.0014	0.6847 $\pm$ 0.0147

This audit separates two effects. KoBBQ itself is condition-separable when labeled KoBBQ examples are available, but an English-BBQ-trained embedding classifier does not transfer directly to Korean representations under the same embedding model. Thus, the weaker KoBBQ end-to-end transfer result in Appendix G is better interpreted as a cross-lingual representation and policy-transfer limitation rather than evidence that KoBBQ lacks an ambiguous/disambiguated boundary.

APPENDIX C RESIDUAL AMBIGUOUS-BIAS COUNTS

Table 12 reports raw ambiguous residual counts for the main predicted-condition variant. The mean residual denominator is only 3.6 non-unknown ambiguous predictions out of 664

ambiguous examples per seed. Thus, a residual stereotype-vs-anti-stereotype score can swing sharply after only one or two examples. This is why the paper does not claim the best residual ambiguous-bias score as a primary result.

TABLE 12. Residual ambiguous predictions for the predicted-condition variant.

Seed	Amb. n	Unknown correct	Residual	Stereo	Anti
42	664	663	1	0	1
123	664	664	0	0	0
456	664	661	3	0	3
789	664	659	5	5	0
999	664	655	9	3	6

APPENDIX D SIGNAL AND SAE AUDIT

Table 13 reports the five-seed signal-masking audit. The magnitudes are small for most individual signals, including s_7 . Table 14 further shows that all-zero, s_3 -only, and four-signal MoE variants remain competitive on the clean split. This supports the conservative interpretation that the dominant clean-split behavior comes from condition-aware selective abstention rather than a single SAE feature or a uniquely necessary seven-signal MoE.

TABLE 13. Signal masking audit over five seeds. Values are mean changes with standard deviations from the artifact builder.

Signal	ΔA_{amb}	ΔA_{dis}	ΔFAR
s_1 evid.	-0.0015 ± 0.0028	-0.0018 ± 0.0049	$+0.0036 \pm 0.0066$
s_2 cf.	-0.0012 ± 0.0007	$+0.0015 \pm 0.0018$	$+0.0012 \pm 0.0039$
s_3 conf.	-0.0012 ± 0.0022	0.0000 ± 0.0034	$+0.0024 \pm 0.0043$
s_4 cons.	-0.0003 ± 0.0020	-0.0015 ± 0.0034	$+0.0033 \pm 0.0055$
s_5 head	-0.0012 ± 0.0020	-0.0009 ± 0.0008	$+0.0009 \pm 0.0020$
s_6 prompt	$+0.0009 \pm 0.0025$	-0.0039 ± 0.0045	$+0.0087 \pm 0.0081$
s_7 SAE	-0.0015 ± 0.0021	-0.0006 ± 0.0051	$+0.0015 \pm 0.0070$

TABLE 14. MoE signal-subset audit over five seeds. The first three reduced variants inherit the same disambiguated keep/override decisions as the condition-only rule, explaining their identical Acc_{dis} and FAR values.

MoE variant	Acc_{amb}	Acc_{dis}	FAR
All signals zeroed	1.0000 ± 0.0000	0.8789 ± 0.0070	0.0726 ± 0.0067
s_3 confidence only	0.9985 ± 0.0015	0.8789 ± 0.0070	0.0726 ± 0.0067
s_1, s_3, s_4, s_6 core	0.9988 ± 0.0027	0.8789 ± 0.0070	0.0726 ± 0.0067
Full seven-signal MoE	0.9946 ± 0.0054	0.8732 ± 0.0108	0.0843 ± 0.0193

APPENDIX E THRESHOLD AUDIT

The low-threshold plateau in Table 5 shows that $\tau_{dis} \in [0.00, 0.05]$ gives comparable behavior. Table 15 reports the seed-level $\tau_{dis} = 0.05$ audit. We interpret this as evidence for a stable low-threshold region, not as a proof of a continuous optimum at a particular threshold.

TABLE 15. Clean test behavior at the audited $\tau_{dis} = 0.05$ setting.

Seed	Acc_{amb}	Acc_{dis}	FAR
42	0.9985	0.8705	0.0934
123	1.0000	0.8810	0.0663
456	0.9955	0.8720	0.0693
789	0.9925	0.8870	0.0768
999	0.9864	0.8584	0.1130

APPENDIX F RULE-BASED EXPLANATION ARTIFACTS

Table 16 summarizes the deterministic explanation labels generated for the predicted-condition variant over five clean

test seeds. Runtime explanations are produced from predicted condition and signal flags only. The audit labels in this table additionally use gold BBQ metadata after evaluation, so they are reported as interpretability artifacts rather than deployment inputs.

TABLE 16. Rule-based explanation audit over 6,640 seed-level decisions.

Decision label	Count	Share
Utility-preserving keep	2899	0.4366
Ambiguous abstention	1858	0.2798
Stereotyped raw answer blocked	995	0.1498
Anti-stereotyped unsupported answer blocked	449	0.0676
False abstention	280	0.0422
Wrong stereotyped keep	76	0.0114
Wrong anti-stereotyped keep	65	0.0098
Anti-stereotype slip	10	0.0015
Stereotype bias slip	8	0.0012

The generated artifact also stores per-instance runtime and audit text. For example, when a raw ambiguous-context answer maps to a stereotyped group and the final answer is unknown, the runtime explanation states that the condition classifier marked the context as ambiguous and that the policy selected the unknown option instead of retaining the raw answer. The benchmark audit label then records that this was a stereotyped raw answer blocked by abstention. This makes the qualitative behavior reproducible without relying on free-form model-generated rationales.

APPENDIX G ADDITIONAL ROBUSTNESS CHECKS

Before the final clean experiment suite, a full v2 five-seed run over 8,864 saved-signal examples achieved $Acc_{amb} = 0.9977 \pm 0.0011$, $Acc_{dis} = 0.8736 \pm 0.0016$, and $FAR = 0.0832 \pm 0.0059$. An additional single full run achieved $Acc_{amb} = 0.9993$, $Acc_{dis} = 0.8748$, and $FAR = 0.0754$. Open-BBQ transfer was rerun under the finalized clean protocol reported in Table 4; an older pre-finalization protocol used the earlier v2 threshold/checkpoint stack before the clean low-threshold and condition-only audits were fixed. In practice, the older variant operated with a more conservative disambiguated-retention setting and did not benefit from the finalized validation-only low-threshold audit, which accounts for its higher FAR. That older variant achieved $Acc_{amb} = 0.9527$, $Acc_{dis} = 0.7939$, and $FAR = 0.1685$ on the same 3,300-example scale and is retained here only as historical protocol sensitivity.

Harder transfer datasets show mixed behavior for the predicted-condition transfer variant. ImplicitBBQ-style transfer achieved $Acc_{amb} = 0.8227$, $Acc_{dis} = 0.5464$, and $FAR = 0.3208$. KoBBQ end-to-end transfer achieved $Acc_{amb} = 0.6557$, $Acc_{dis} = 0.6475$, and $FAR = 0.2186$. This weaker KoBBQ end-to-end result is consistent with Appendix B: English-to-KoBBQ condition prediction is near random even though within-KoBBQ condition prediction is nearly perfect. KoBBQ is therefore especially important as a limitation because the task is BBQ-like but the language and representation distribution differ; weaker performance there suggests that English BBQ condition separability and LlamaScope SAE features do not automatically transfer. StereoSet

and WinoGender do not match the BBQ abstention formulation closely, so we keep them as qualitative robustness checks rather than main evidence.

APPENDIX H REPRODUCIBILITY AND ARTIFACT CONTENTS

The clean experiments use seeds 42, 123, 456, 789, and 999. The supplementary artifact contains saved predictions, raw metrics, paired tests, threshold audits, residual-bias counts, low-label hybrid summaries, rule-based explanation artifacts, and report-building scripts. The main entry points are `scripts/run_clean_experiments.py` for the clean suite, `scripts/run_hybrid_abstention_audits.py` for low-label and uncertainty fallback audits, `scripts/generate_rule_based_explanations.py` for deterministic explanations, `scripts/run_transfer_condition_audits.py` for KoBBQ condition-transfer audits, `scripts/run_loco_clean.py` for LOCO, `src.transfer.run_open_bbq` for Open-BBQ transfer, `src.analysis.multi_seed` for cross-LLM summaries, and the report-building script for manuscript tables. The supplementary `reproducibility.md` file gives the exact command-line flags and output directories. These entries are included to make the clean experiment suite auditable rather than relying only on summary tables in the manuscript.

ACKNOWLEDGMENT

The author thanks the maintainers of the BBQ benchmark and open model/tooling ecosystems used in this work. OpenAI's GPT-based coding and writing assistant, Codex, was used for drafting, editing, coding, and organizing parts of this manuscript and repository; the author reviewed, corrected, and validated the final content, experiments, and claims.

REFERENCES

- [1] A. Parrish, A. Chen, N. Nangia, V. Padmakumar, J. Phang, J. Thompson, P. M. Htut, and S. R. Bowman, "BBQ: A hand-built bias benchmark for question answering," in *Findings of the Association for Computational Linguistics: ACL 2022*. Association for Computational Linguistics, 2022, pp. 2086–2105. [Online]. Available: <https://aclanthology.org/2022.findings-acl.165/>
- [2] S. Bae, Y. Choi, and J.-H. Lee, "DeCAP: Context-adaptive prompt generation for debiasing zero-shot question answering in large language models," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.19426>
- [3] I. O. Gallegos, R. A. Rossi, J. Barrow, M. M. Tanjim, T. Yu, H. Deilamsalehy, R. Zhang, S. Kim, and F. Dernoncourt, "Self-debiasing large language models: Zero-shot recognition and reduction of stereotypes," *arXiv preprint arXiv:2402.01981*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.01981>
- [4] Y. Li, Z. Fan, R. Chen, X. Gai, L. Gong, Y. Zhang, and Z. Liu, "FairSteer: Inference time debiasing for large language models with dynamic activation steering," in *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. [Online]. Available: <https://arxiv.org/abs/2504.14492>
- [5] C. K. Chow, "On optimum recognition error and reject tradeoff," *IEEE Transactions on Information Theory*, vol. 16, no. 1, pp. 41–46, 1970.
- [6] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: <https://papers.nips.cc/paper/7073-selective-classification-for-deep-neural-networks>
- [7] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, ser. *Proceedings of Machine Learning Research*, vol. 70. PMLR, 2017, pp. 1321–1330. [Online]. Available: <https://proceedings.mlr.press/v70/guo17a.html>
- [8] T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. L. Turner, C. Anil, C. Denison, A. Askell, R. Lasenby, Y. Wu, S. Kravec, N. Schiefer, T. Maxwell, N. Joseph, Z. Hatfield-Dodds, A. Tamkin, K. Nguyen, B. McLean, J. E. Burke, T. Hume, S. Carter, T. Henighan, and C. Olah, "Towards monosemanticity: Decomposing language models with dictionary learning," *Transformer Circuits Thread*, 2023. [Online]. Available: <https://transformer-circuits.pub/2023/monosemantic-features/>
- [9] Z. He, W. Shu, X. Ge, L. Chen, J. Wang, Y. Zhou, F. Liu, Q. Guo, X. Huang, Z. Wu, Y.-G. Jiang, and X. Qiu, "Llama scope: Extracting millions of features from llama-3.1-8b with sparse autoencoders," *arXiv preprint arXiv:2410.20526*, 2024. [Online]. Available: <https://arxiv.org/abs/2410.20526>
- [10] M. Nadeem, A. Bethke, and S. Reddy, "StereoSet: Measuring stereotypical bias in pretrained language models," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*. Association for Computational Linguistics, 2021, pp. 5356–5371. [Online]. Available: <https://aclanthology.org/2021.acl-long.416/>
- [11] N. Nangia, C. Vania, R. Bhalerao, and S. R. Bowman, "CrowS-Pairs: A challenge dataset for measuring social biases in masked language models," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2020, pp. 1953–1967. [Online]. Available: <https://aclanthology.org/2020.emnlp-main.154/>
- [12] R. Rudinger, J. Naradowsky, B. Leonard, and B. Van Durme, "Gender bias in coreference resolution," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics*. Association for Computational Linguistics, 2018, pp. 8–14. [Online]. Available: <https://aclanthology.org/N18-2002/>
- [13] AI at Meta, "The Llama 3.1 herd of models," Model card and technical report, 2024. [Online]. Available: <https://github.com/meta-llama/llama-models>
- [14] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, H. Lin, J. Yang, J. Tu, J. Zhang, J. Yang, J. Yang, J. Zhou, J. Lin, K. Dang, K. Lu, K. Bao, K. Yang, L. Yu, M. Li, M. Xue, P. Zhang, Q. Zhu, R. Men, R. Lin, T. Li, T. Tang, T. Xia, X. Ren, X. Ren, Y. Fan, Y. Su, Y. Zhang, Y. Wan, Y. Liu, Z. Cui, Z. Zhang, and Z. Qiu, "Qwen2.5 technical report," *arXiv preprint arXiv:2412.15115*, 2024. [Online]. Available: <https://arxiv.org/abs/2412.15115>
- [15] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. Le Scao, T. Lavril, T. Wang, T. Lacroix, and W. El Sayed, "Mistral 7b," *arXiv preprint arXiv:2310.06825*, 2023. [Online]. Available: <https://arxiv.org/abs/2310.06825>

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